SILVER OAK COLLEGE OF COMPUTER APPLICATION

Department of Computer Application

Bachelor of Computer Application — Semester 4

Project Documentation

College Complaint & Feedback Management System

Course: Integrated Web Solutions with Django and Basics of Odoo

Course Code: 4040233205

Academic Year: 2025-26

Group Members

Sr.	Name	Enrollment Number	Role
1	Solanki Yash Ashvin Kumar	2404030100391	Django Development
2	Patel Manan Vipul Kumar	2404030100329	Django Templates
3	Prajapati Jaydeep Ramji Bhai	2404030100363	Odoo Module
4	Tiwari Shani Vijay Bhai	2404030101039	Documentation

1. Problem Definition

In a college environment, students often face various issues related to faculty, infrastructure, canteen, library, and other facilities. However, there is no structured system for students to formally raise these concerns. Complaints are often made verbally or informally, which leads to no proper tracking or resolution.

This project addresses the need for a digital, structured, and transparent Complaint & Feedback Management System where students can submit complaints, and administrators can view, manage, and resolve them efficiently.

2. Objective of Application

- To provide students with a simple and easy-to-use platform to submit complaints.
- To allow administrators to view, filter, update and resolve complaints.
- To maintain a proper record of all complaints with status tracking.
- To implement Django concepts including ORM, Forms, Templates, and CRUD operations.
- To develop an Odoo custom module for complaint management.
- To improve transparency and accountability in college complaint resolution.

3. Key Features of Application

Django Application Features

- Student complaint submission form with validation
- View all complaints in a tabular format
- Filter complaints by status (Pending / In Progress / Resolved)
- Admin can update complaint status
- Delete complaint functionality
- Statistics dashboard on home page (Total, Pending, Resolved)
- Template inheritance using base.html
- Responsive UI using Bootstrap 5

Odoo Module Features

- Custom Odoo module: complaint_management
- Model with fields: student name, department, category, description, status
- Many2one relational field linking Complaint to Department
- Form view and Tree (list) view

- Custom menu and actions
- ORM operations: create, write, search, unlink

4. System Architecture

The application follows the Django MVT (Model-View-Template) architecture:

Layer	Component	Description
Model	models.py	Defines Department and Complaint database tables
View	views.py	Contains business logic and query handling
Template	HTML Files	Renders data in browser using Bootstrap 5
Form	forms.py	Handles input validation and form rendering
URL	urls.py	Maps URLs to appropriate view functions
Admin	admin.py	Django admin panel configuration

User Flow: Student visits homepage → Fills complaint form → Form validated → Data saved to SQLite DB → Admin views complaints → Admin updates status → Student can check status

5. Django Modules and Functionalities Used

Concept	File	Usage
Project & App Creation	manage.py	Created complaint_system project and complaints app
URL Mapping	urls.py	Mapped 6 URL patterns to views
HTML Templates	base.html + others	Template inheritance with {% extends %} and {% block %}
Django Forms	forms.py	ModelForm with custom validation
CRUD Operations	views.py	Create, Read, Update, Delete complaints using ORM
GET & POST Methods	views.py	GET for displaying forms, POST for submitting data
Django ORM	views.py	filter(), count(), order_by(), save(), delete()
Django Admin	admin.py	Registered models with list_display and filters

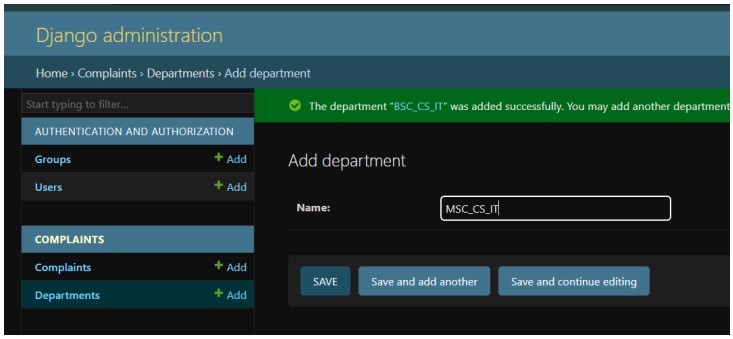
6. Odoo Modules and Functionalities Used

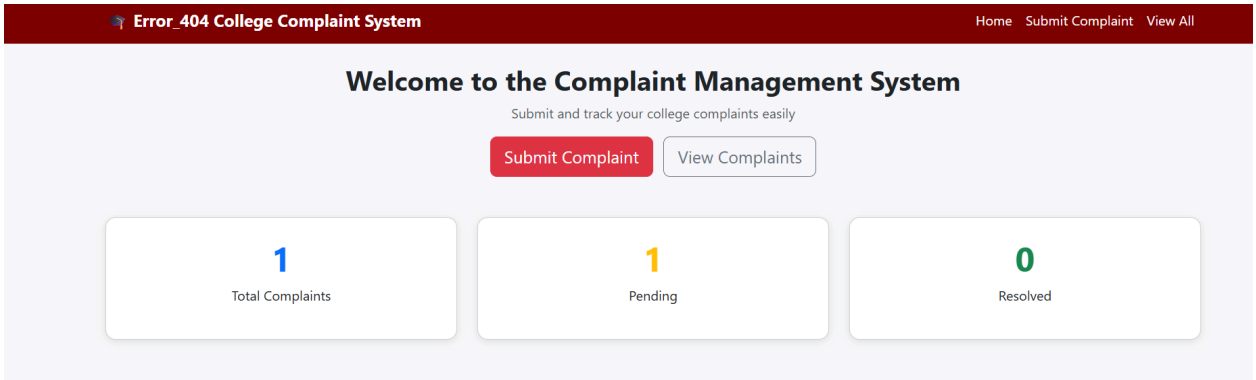
Concept	File	Usage
Custom Module	__manifest__.py	Created complaint_management Odoo module
Model Definition	models.py	Defined fields: name, department_id, category, status
Relational Field	models.py	Many2one field linking Complaint to Department
Form View	views.xml	Form view for adding/editing complaints
Tree View	views.xml	List view showing all complaints
Menu & Action	views.xml	Custom menu item and window action
ORM Methods	models.py	create, write, search, unlink operations

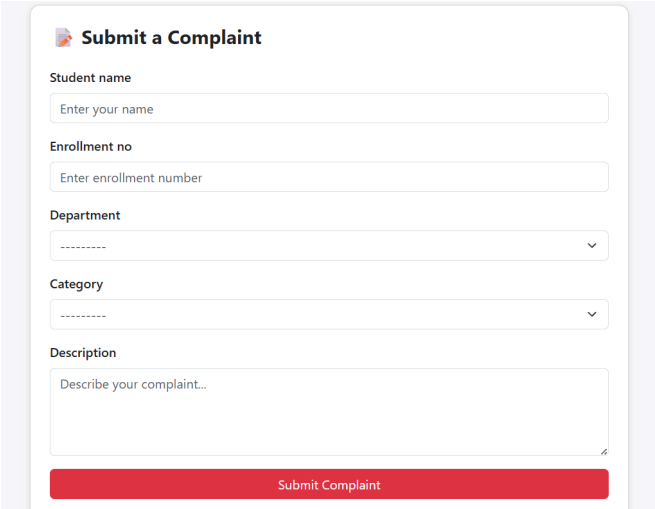
7. Tools and Technologies Used

Technology	Version	Purpose
Python	3.11	Backend programming language
Django	5.2	Web framework for application development
SQLite	3.x	Default database for Django development
Bootstrap	5.3	Frontend CSS framework for responsive UI
Odoo	16 Community	ERP platform for custom module development
VS Code	Latest	Code editor
Git	Latest	Version control
Windows	11	Operating System

8. Application Screenshots

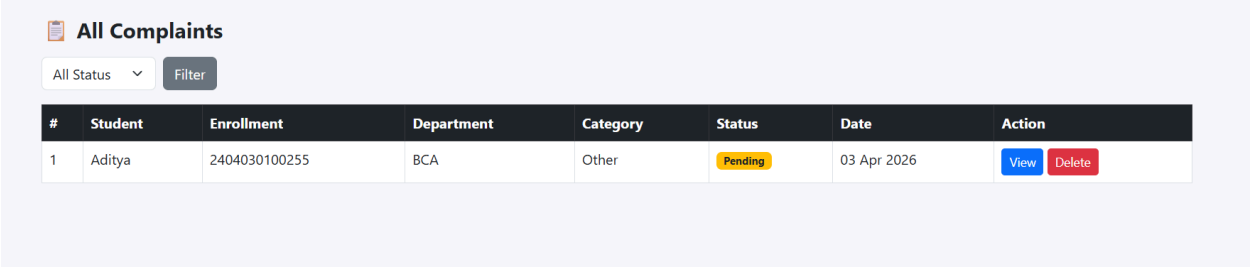



Error_404 College Complaint System
Home
Submit Complaint
View All
Welcome to the Complaint Management System
Submit and track your college complaints easily
Submit Complaint
View Complaints
1
Total Complaints
1
Pending
0
Resolved


Submit a Complaint
Student name
Enter your name
Enrollment no
Enter enrollment number
Department

Category

Description
Describe your complaint...
Submit Complaint


All Complaints
All Status
Filter
Student Enrollment Department Category Status Date Action
1 Aditya 2404030100255 BCA Other Pending 03 Apr 2026 View Delete

9. Learning Outcomes

- Understood how to create and structure a Django project and application.
- Learned URL routing and mapping using urls.py.
- Implemented Django ORM for database operations without writing raw SQL.
- Created Django Forms with built-in and custom validations.
- Applied template inheritance to avoid code repetition.
- Handled GET and POST HTTP methods in Django views.
- Built a custom Odoo module from scratch with models, views, and menus.
- Used relational fields (Many2one) in Odoo to link models.
- Gained hands-on experience with full-stack web development.

10. Conclusion

The College Complaint & Feedback Management System successfully demonstrates the practical application of Django and Odoo concepts learned in the course. The project covers all mandatory syllabus topics including Django ORM, Forms, Templates, URL mapping, CRUD operations, and Odoo custom module development with relational fields and views.

The application provides a clean, user-friendly interface for students to submit complaints and for administrators to manage and resolve them. The system improves transparency in the complaint resolution process and serves as a practical mini-project that can be further extended with features like email notifications, user authentication, and analytics.

This project helped the team gain hands-on experience in full-stack web development using modern frameworks and reinforced the importance of proper database design, logical application flow, and clean code practices.

11. Evaluation Criteria

Description	Marks	Status
Title & Topic Selection	5	✓ Completed
Proper Database Design	5	✓ Completed
Logical Application Flow	5	✓ Completed
Viva	5	Pending
Documentation	5	✓ Completed
Total Marks	25	